NMR Spectroscopy

Introduction

In 1946, NMR was co-discovered by Purcell, Pound and Torrey of Harvard University and Bloch, Hansen and Packard of Stanford University.

The discovery first came about when it was noticed that **magnetic nuclei**, such as ^{1}H and ^{31}P (read: proton and Phosphorus 31) were able to **absorb radio frequency energy** when placed in a **magnetic field of a strength that was specific to the nucleus**. Upon absorption, the nuclei begin to **resonate** and different atoms within a molecule resonated at different frequencies.

This observation allowed a detailed analysis of the structure of a molecule. Since then, NMR has been applied to solids, liquids and gasses, kinetic and structural studies, resulting in **6 Nobel prizes** being awarded in the field of NMR.

Similar to the UV and IR spectroscopy, **nuclear magnetic resonance (NMR) spectroscopy is also an absorption spectroscopy** in which samples absorb electromagnetic radiation in the **radio-frequency** region (3 MHz to 30,000 MHz) at frequencies governed by the characteristics of the sample.

NMR Spectroscopy

- ***Nuclear magnetic resonance*** **(NMR) spectroscopy** helps to identify the ***carbon-hydrogen framework*** of an organic compound.

- **Nuclei** that have an **odd number of protons** or an **odd number of neutrons** (**or both**) have a property called **spin** that allows them (**^{1}H, ^{13}C, ^{15}N, ^{19}F,** and **^{31}P**) to be studied by NMR.

- **Nuclei** such as **^{12}C** and **^{16}O** do ***not*** have **spin** and therefore cannot be studied by NMR.

- Because hydrogen nuclei (protons) were the first nuclei studied by NMR, the acronym ***NMR*** is generally assumed to mean **^{1}H NMR (proton magnetic resonance)** .

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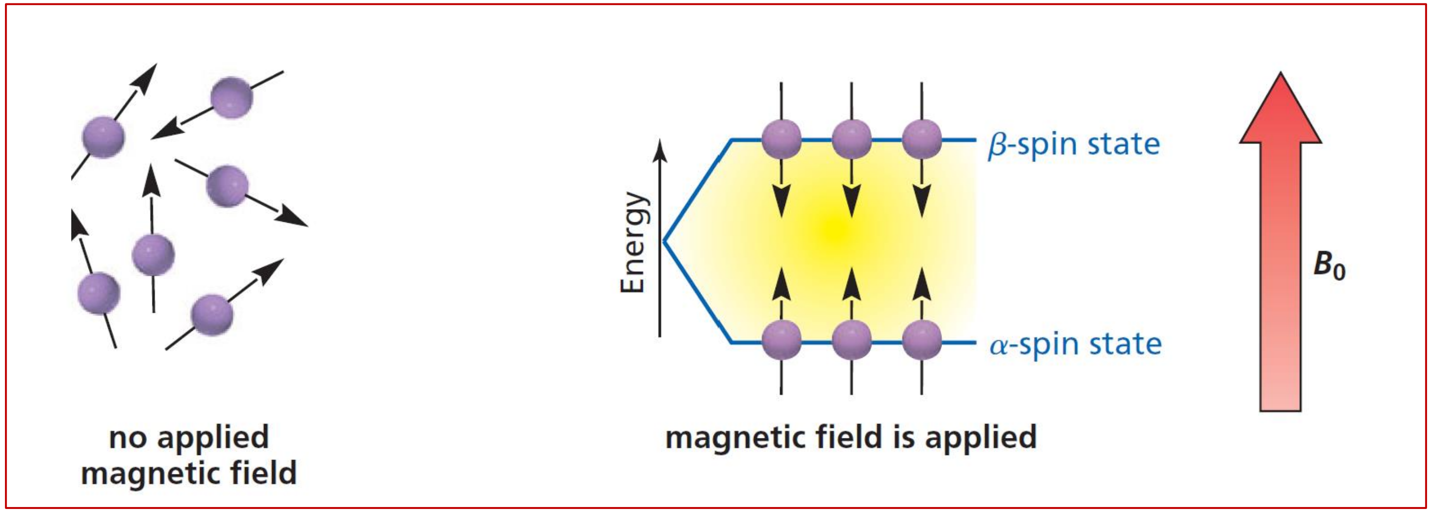
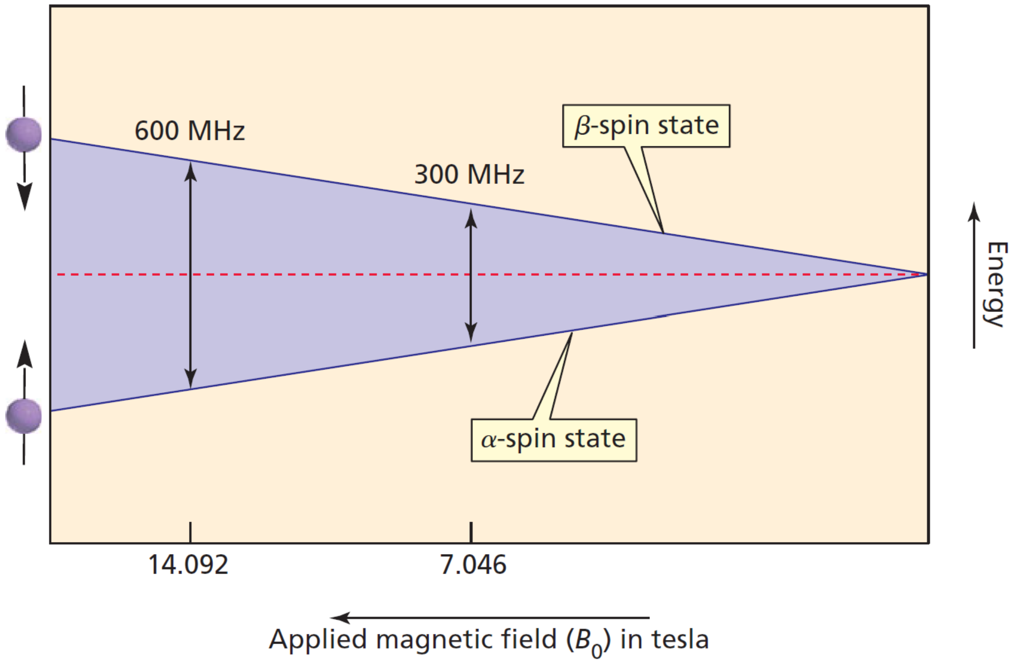
- As a result of its charge, **a nucleus** with **spin** has a **magnetic moment** and generates a **magnetic field** similar to the magnetic field generated by a small bar magnet. In the absence of an applied magnetic field, the magnetic moments of the nuclei are randomly oriented.
- However, when placed between the poles of a strong magnet, the magnetic moments of the nuclei align either ***with* or *against*** the applied magnetic field



- Nuclei with magnetic moments that align with the field are in the lower-energy **α-spin state,** whereas those with magnetic moments that align against the field are in the higher energy **β-spin state.**
- The **β**-spin state is higher in energy because more energy is needed to align the magnetic moments against the field than with it. As a result, more nuclei are in the **α**-spin state.
- The difference in the populations is very small (about 20 out of 1 million protons), but it is sufficient to form the basis of NMR spectroscopy.

NMR Spectroscopy

The energy difference (E) between the **α**- and **β**-spin states depends on the strength of the **applied magnetic field** (B_0): the greater the strength of the applied magnetic field, the greater the E



When a sample is subjected to a pulse of radiation whose energy corresponds to the difference in energy (E) between the **α**- and **β**-spin states, nuclei in the **α** -spin state are promoted to the **β**-spin state. This transition is called "*flipping*" the spin.

The radiation used to supply this energy is in the ***radio frequency*** (rf) region of the electromagnetic spectrum and is called **rf radiation**.

NMR Spectroscopy

- When the nuclei absorb **rf radiation** and flip their spins, they generate **signals** whose frequency depends on the difference in energy (*E*) between the **α-** and **β**-spin states.
- The NMR spectrometer **detects these signals** and plots their frequency versus their intensity; this plot is an NMR spectrum.
- The nuclei are said to be ***in resonance*** with the **rf radiation**, hence the term **nuclear magnetic resonance.**
- In this context, **"resonance"** refers to the nuclei flipping back and forth between the **α-** and **β** -spin states in response to the rf radiation; it has nothing to do with the "resonance" associated with electron delocalization.

The following equation shows that the energy difference between the spin **states (*E*) depends on the operating frequency of the spectrometer (ν)** , which depends, in turn, on the strength of the magnetic field **(B_0),** measured in tesla **(T)**, and the *gyromagnetic ratio* (**γ**) ; *h* is Planck's constant

$$\boxed{\Delta E = h\nu = h\frac{\gamma}{2\pi}B_0}$$

The **gyromagnetic ratio** is a constant that depends on the particular kind of nucleus. In the case of a proton, **γ** = 2.675 $\times 10^8\ T^{-1}s^{-1}$; in the case of a ^{13}C nucleus, it is 6.688 x $10^7\ T^{-1}s^{-1}$. Cancelling Planck's constant on both sides of the equation gives

$$\boxed{\nu = \frac{\gamma}{2\pi}B_0}$$

$$\boxed{\gamma = \frac{q}{2m}}$$

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The following calculation shows that if an $_1$H NMR spectrometer is equipped with a magnet that generates a magnetic field of 7.046 T, then the spectrometer will require an operating frequency of 300 MHz (megahertz):

$$\begin{aligned} v &= \frac{\gamma}{2\pi} B_0 \\ &= \frac{2.675 \times 10^8}{2(3.1416)} \mathrm{T^{-1}s^{-1}} \times 7.046\ \mathrm{T} \\ &= 300 \times 10^6\ \mathrm{Hz} = 300\ \mathrm{MHz} \end{aligned}$$

The equation shows that the strength of the *magnetic field* (B_0) *is proportional to the operating frequency* (MHz) of the spectrometer. Therefore, if the spectrometer has **a more powerful magnet**, then it must have a higher **operating frequency.** For example, a magnetic field of 14.092 T requires an operating frequency of 600 MHz.

Because each kind of nucleus has its **own gyromagnetic ratio**, different frequencies are required to bring different kinds of nuclei into resonance. For example, an NMR spectrometer that requires a frequency of **300 MHz to flip the spin of an ^{1}H nucleus** requires a frequency of **75 MHz to flip the spin of a ^{13}C nucleus**. NMR spectrometers are equipped with radiation sources that **can be tuned to different frequencies** so that they can be used to obtain NMR spectra of different kinds of nuclei (^{1}H, ^{13}C, ^{15}N, ^{19}F, ^{31}P).

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Nuclei	Spin	Gyromagetic Ratio (MHz/T)	Natural Abundance (%)
^{1}H	1/2	42.576	99.9985
^{13}C	1/2	10.705	1.07
^{31}P	1/2	17.235	100

Nuclear spin may be related to the nucleon composition of a nucleus in the following manner:

Odd mass nuclei (i.e. those having an odd number of nucleons) have fractional spins. Examples are I = 1/2 (^{1}H, ^{13}C, ^{19}F), I = 3/2 (^{11}B) & I = 5/2 (^{17}O).

Even mass nuclei composed of odd numbers of protons and neutrons have integral spins. Examples are I = 1 (^{2}H, ^{14}N).

Even mass nuclei composed of even numbers of protons and neutrons have zero spin (I = 0). Examples are ^{12}C, and ^{16}O

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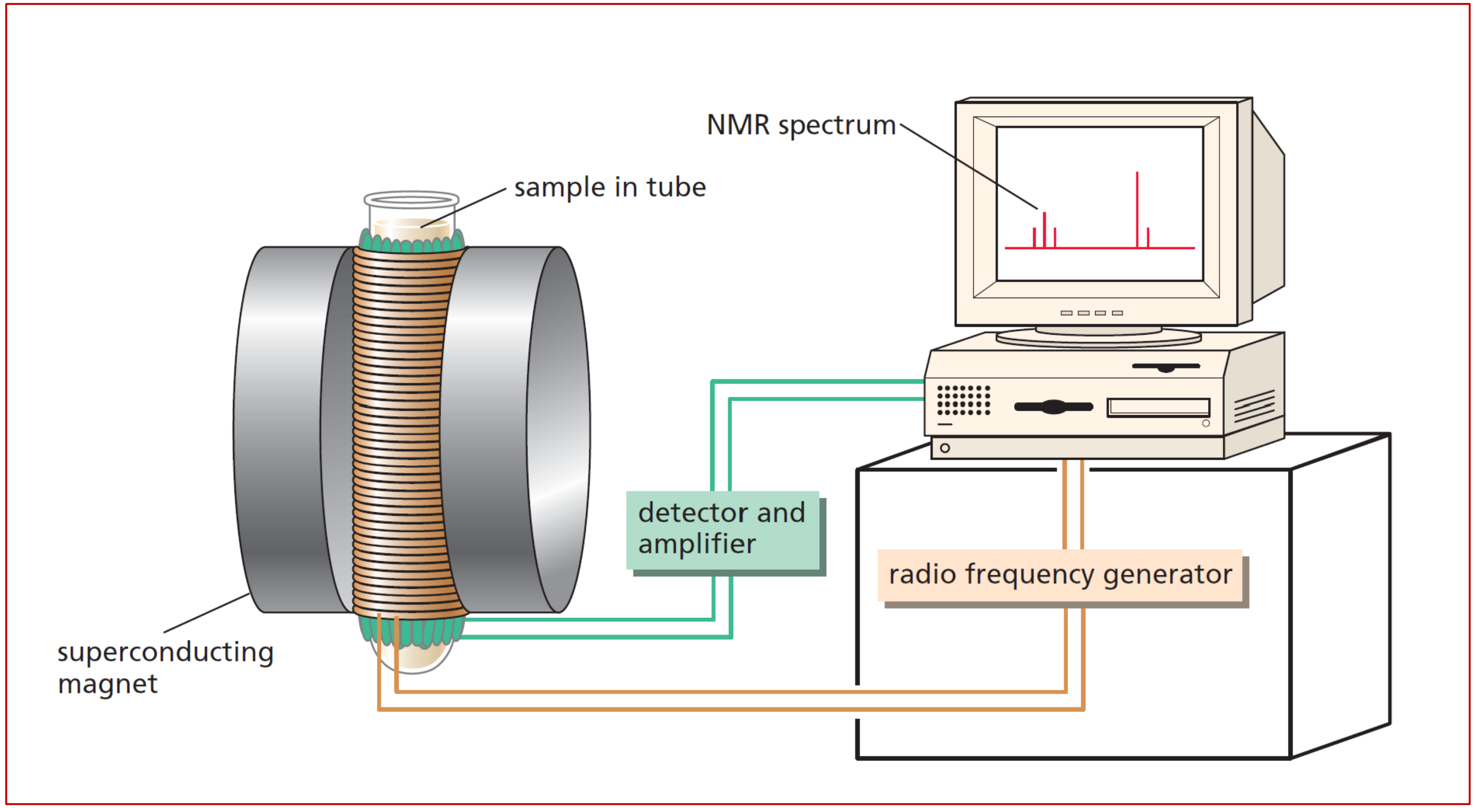
To obtain an **NMR spectrum,** a small amount of a compound is dissolved in about 0.5 mL of solvent. This solution is put into a long, thin glass tube, which is then placed within a powerful magnetic field. **Spinning the sample tube about its long axis** averages the position of the molecules in the magnetic field, which **increases the resolution of the spectrum.**



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SHIELDING CAUSES DIFFERENT HYDROGENS TO SHOW SIGNALS AT DIFFERENT FREQUENCIES

A nucleus, is embedded in a cloud of electrons that partly ***shields*** it from the applied magnetic field. Fortunately for chemists, the **shielding** varies for different hydrogens in a molecule. **In other words, all the hydrogens do not experience the same magnetic field.**

What causes shielding?

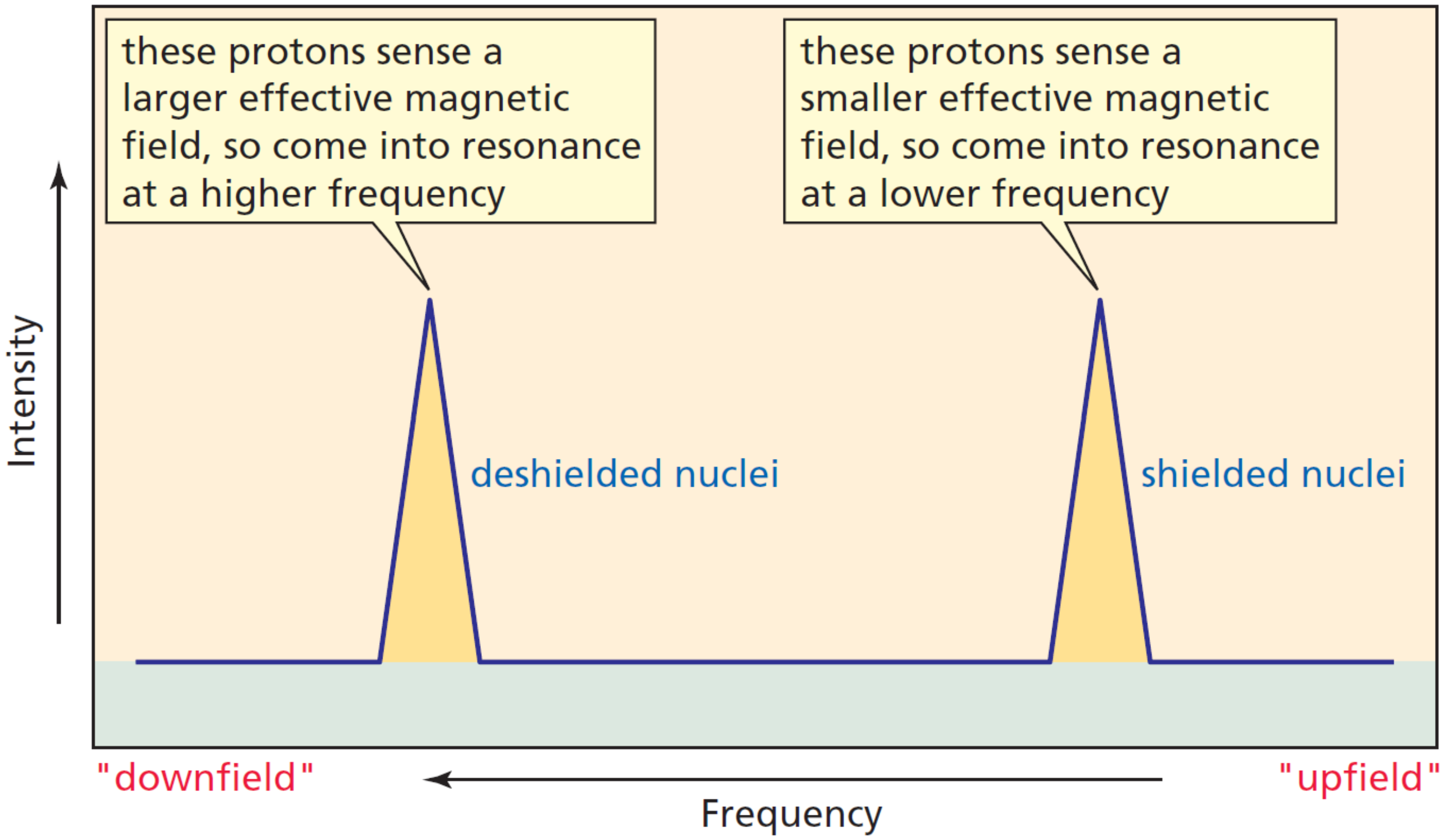
In a magnetic field, the **electrons circulate about the nuclei and induce a local magnetic field** that acts in opposition to the applied magnetic field and, therefore, subtracts from it. As a result, the **effective magnetic field** —the amount of magnetic field that the nuclei actually **"sense"** through the surrounding electrons—is somewhat **smaller than the applied magnetic field**:

$$B_{\text{effective}} = B_{\text{applied}} - B_{\text{local}}$$

This means that the greater the electron density of the environment in which the proton is located, the more the proton is shielded from the applied magnetic field and the greater is B_{local}. This type of shielding is called **diamagnetic shielding .**

NMR Spectroscopy

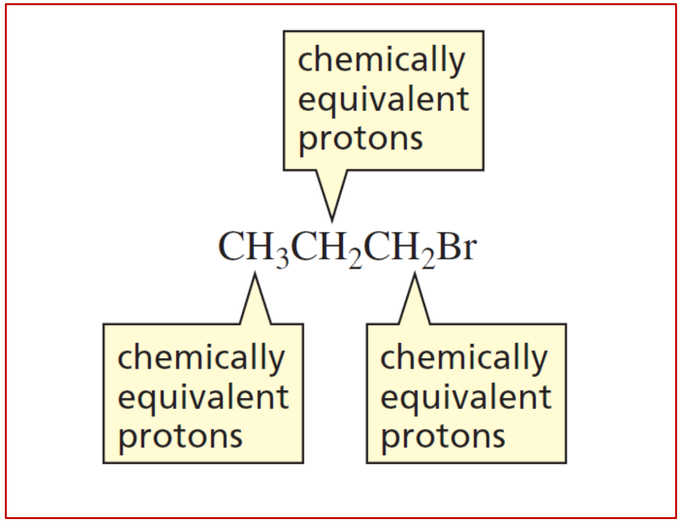
- Thus, protons in electron-rich environments sense a ***smaller effective magnetic field***. Therefore, they require a ***lower frequency*** to come into resonance—that is, flip their spin—because *E* is smaller.
- Protons in electron-poor environments sense **a *larger effective magnetic field*** and so require a ***higher frequency*** to come into resonance because *E* is larger.



NMR Spectroscopy

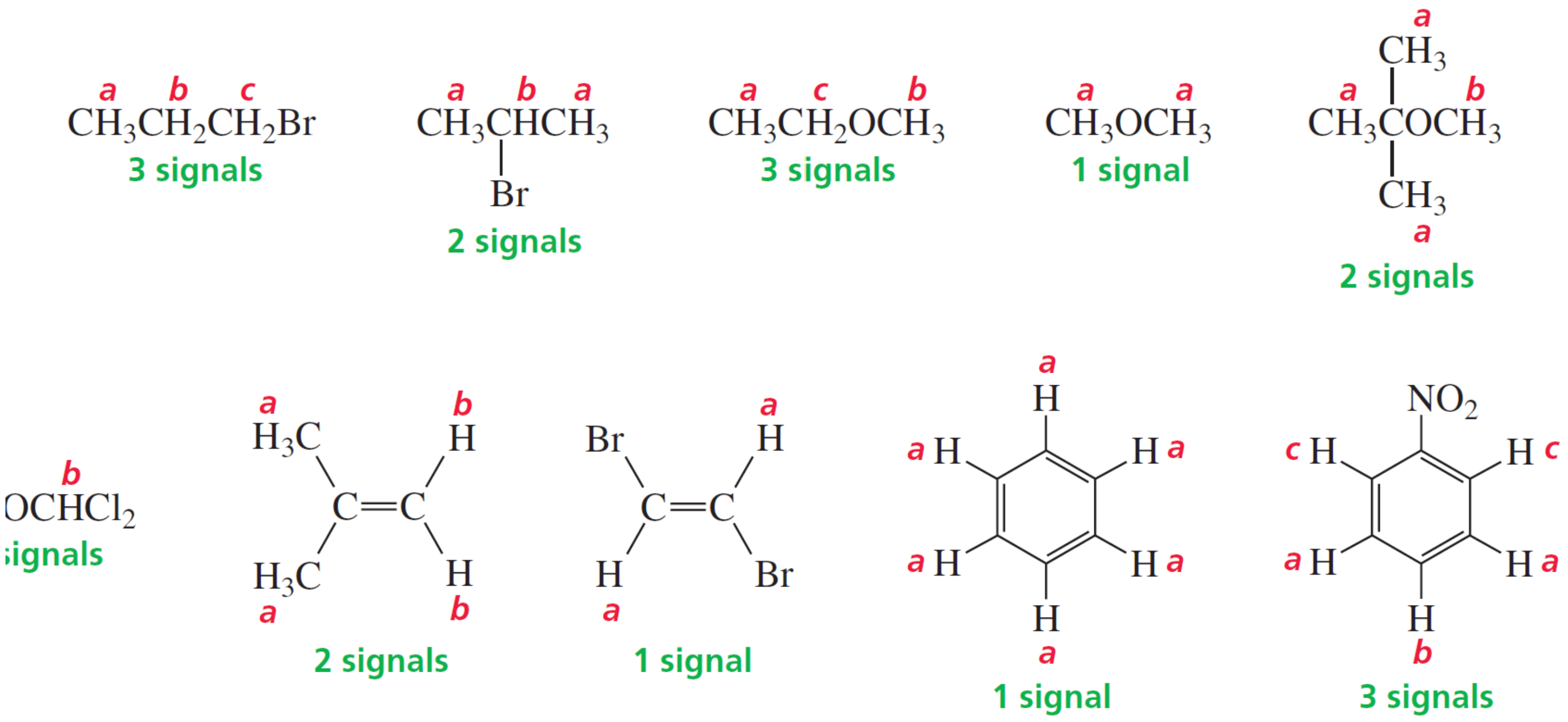
THE NUMBER OF SIGNALS IN AN ^{1}H NMR SPECTRUM

Protons in the same environment are called **chemically equivalent protons.** For example, **1-bromopropane has three different** sets of chemically equivalent protons: the three methyl protons are chemically equivalent because of rotation about the C-C bond; the two methylene (CH_2) protons on the middle carbon are chemically equivalent; and the two methylene protons on the carbon bonded to the bromine make up the third set of chemically equivalent protons.



Each set of chemically equivalent protons in a compound produces a separate signal in its ^{1}H NMR spectrum. Thus, 1-bromopropane has three signals in its ^{1}H NMR spectrum because it has three sets of chemically equivalent protons.

2-Bromopropane has two sets of chemically equivalent protons, so it has two signals in its ^1H NMR spectrum. The six methyl protons are equivalent so they produce only one signal, and the hydrogen bonded to the middle carbon gives the second signal.



Ethyl methyl ether has three sets of chemically equivalent protons: the methyl protons on the carbon adjacent to the oxygen, the methylene protons on the carbon adjacent to the oxygen, and the methyl protons on the carbon that is one carbon removed from the oxygen. The chemically equivalent protons in the compounds shown here are designated by the same letter.

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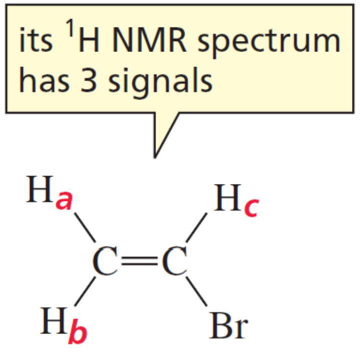
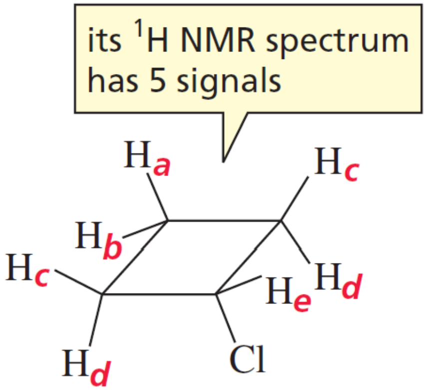
If bonds are prevented from freely rotating, as in a compound with a double bond or in a cyclic compound, two protons on the same carbon may not be equivalent. For example, the H_a and H_b protons of bromoethene are not equivalent because they are not in the same environment: H_a is trans to Br, whereas H_b is cis to Br. Thus, the ^1H NMR spectrum of bromoethene has three signals.



bromoethene
H_a and H_b are not equivalent



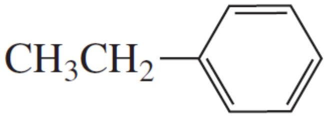
chlorocyclobutane
H_a and H_b are not equivalent
H_c and H_d are not equivalent

The ^1H NMR spectrum of chlorocyclobutane has five signals. The H_a and H_b protons are not equivalent: H_a is trans to Cl, whereas H_b is cis to Cl. Similarly, the H_c and H_d protons are not equivalent.

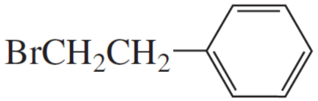
NMR Spectroscopy

Determining the Number of Signals in an ^{1}H NMR Spectrum

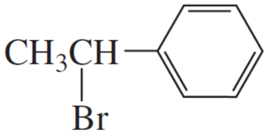
How many signals would you expect to see in the ^{1}H NMR spectrum of ethylbenzene?



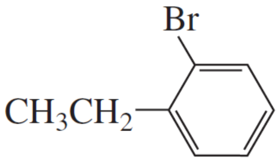
To determine the number of signals you would expect to see in the spectrum, replace each hydrogen in turn by another atom (here we use Br) and name the resulting compound. The number of different names corresponds to the number of signals in the ^{1}H NMR spectrum. We get five different names for the bromosubstituted ethylbenzenes, so we expect to see five signals in the ^{1}H NMR spectrum of ethylbenzene.



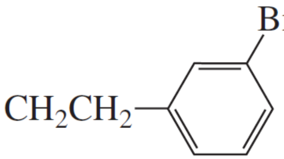
1-bromo-2-phenylethane



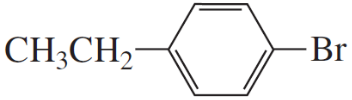
1-bromo-1-phenylethane



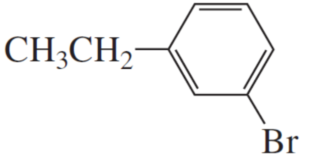
1-bromo-2-ethylbenzene



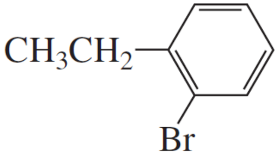
1-bromo-3-ethylbenzene



1-bromo-4-ethylbenzene



1-bromo-3-ethylbenzene



1-bromo-2-ethylbenzene

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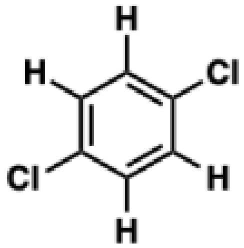
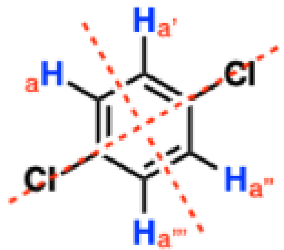
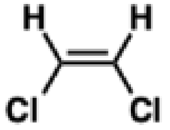
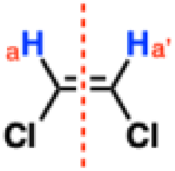
How many ^{1}H NMR signals would these compounds produce?

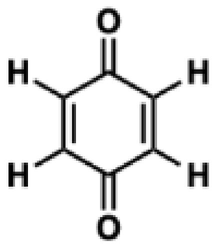
Answer: <u>one signal!</u>

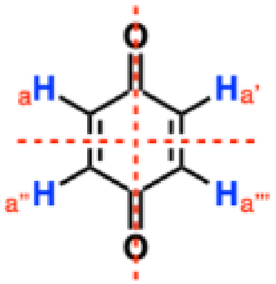






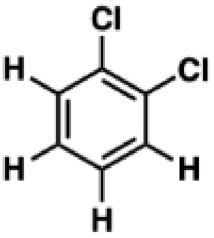


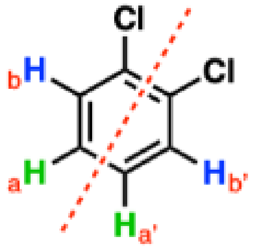




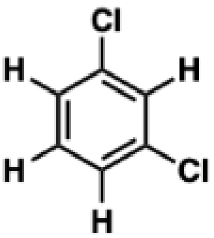
How many ^{1}H NMR signals would these compounds produce?

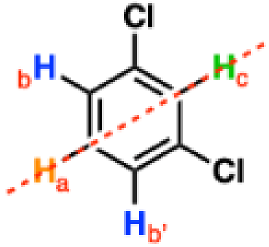
Answer:





<u>2 signals</u>



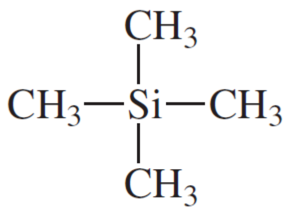


<u>3 signals</u>

NMR Spectroscopy

THE CHEMICAL SHIFT TELLS HOW FAR THE SIGNAL IS FROM THE REFERENCE SIGNAL

A small amount of an inert reference compound is added to the sample tube containing the compound whose NMR spectrum is to be taken. The most commonly used reference compound is **tetramethylsilane (TMS).**



tetramethylsilane
TMS

The methyl protons of TMS are in a more electron-rich environment than are most protons in organic molecules **because silicon is less electronegative than carbon** (their electronegativities are 1.8 and 2.5, respectively). Consequently, the signal for the **methyl protons of TMS is at a lower frequency than most other signals** (that is, the TMS signal appears to the right of the other signals).

- The position at which a signal occurs in an NMR spectrum is called the **chemical shift**. The **chemical shift** is a measure of **how far the signal is from the signal** for the reference compound.
- The most common scale for chemical shifts is the **δ** (delta) scale. The TMS signal defines the zero position on the **δ scale.**

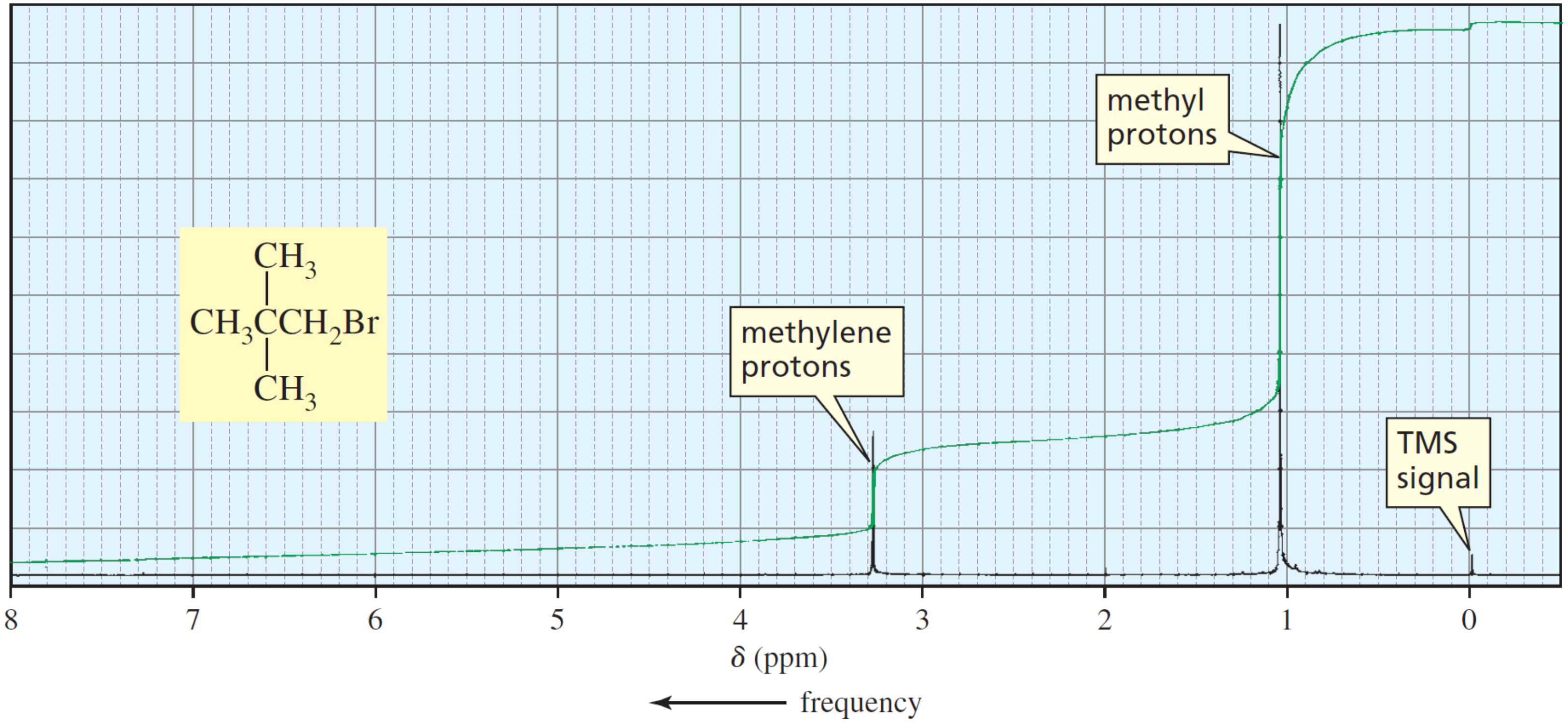
The chemical shift is determined by measuring the distance from the **TMS peak in hertz** and **dividing by the operating frequency of the instrument in megahertz**. Because the units are Hz/MHz, a chemical shift has units of parts per million (ppm) of the operating frequency:

$$\boxed{\delta = \text{chemical shift (ppm)} = \frac{\text{distance downfield from TMS (Hz)}}{\text{operating frequency of the spectrometer (MHz)}}}$$

NMR Spectroscopy

THE CHEMICAL SHIFT TELLS HOW FAR THE SIGNAL IS FROM THE REFERENCE SIGNAL

The ^{1}H NMR spectrum in Figure shows that the chemical shift (**δ**) of the methyl protons is at 1.05 ppm and the chemical shift of the methylene protons, which are deshielded by the electron-withdrawing bromine, is at 3.28 ppm. *Notice that **low- frequency (shielded)** signals have **small δ** (ppm) values, whereas **high-frequency (deshielded)** signals have **large δ** values.*



Figure

The ^{1}H NMR spectrum of 1-bromo-2,2-dimethylpropane. The TMS signal is a reference signal from which chemical shifts are measured; it defines the zero position on the scale.

NMR Spectroscopy

The advantage of the δ scale is that the chemical shift is *independent of the operating frequency of the NMR spectrometer*. Thus, the chemical shift of the methyl protons of 1-bromo-2,2-dimethylpropane is at 1.05 ppm in both a 300-MHz and a 500-MHz instrument. If the chemical shift were reported in hertz instead, it would be at 315 Hz in a 300-MHz instrument and at 525 Hz in a 500-MHz instrument (315/300 = 1.05, 525/500 = 1.05).

The following diagram will help you keep track of the terms associated with NMR spectroscopy:

protons in electron-poor environments	protons in electron-dense environments
deshielded protons	shielded protons
downfield	upfield
high frequency	low frequency
large δ values	small δ values

$\longleftarrow$ δ ppm

$\longleftarrow$ frequency